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$$f(x, y) = x^3 + x^2y^3 - 3y^4 \text{ in } (1, -1), \text{ graad } 2$$

$$f(1, -1) = 1 - 1 - 3 = -3$$

$$\frac{\partial f}{\partial x}(x, y) = 3x^2 + 2xy^3 \Rightarrow \frac{\partial f}{\partial x}(1, -1) = 1$$

$$\frac{\partial f}{\partial y}(x, y) = 3x^2y^2 - 12y^3 \Rightarrow \frac{\partial f}{\partial y}(1, -1) = 15$$

$$\frac{\partial^2 f}{\partial x^2}(x, y) = 6x + 2y^3 \Rightarrow \frac{\partial^2 f}{\partial x^2}(1, -1) = 4$$

$$\frac{\partial^2 f}{\partial x \partial y}(x, y) = 6xy^2 \Rightarrow \frac{\partial^2 f}{\partial x \partial y}(1, -1) = 6$$

$$\frac{\partial^2 f}{\partial y^2}(x, y) = 6x^2y - 36y^2 \Rightarrow \frac{\partial^2 f}{\partial y^2}(1, -1) = -42$$

$$T_2(f, (1, -1))(1 + h, -1 + k) = -3 + h + 15k + \frac{1}{2!} (4h^2 - 12hk - 42k^2) \text{ met } h = x - 1 \text{ en } k = y + 1$$

$$T_2(f, (1, -1))(x, y) = -3 + (x-1) + 15(y+1) + \frac{1}{2!} (4(x-1)^2 - 12(x-1)(y+1) - 42(y+1)^2)$$

$$= -3 + x - 1 + 15y + 15 + 2x^2 - 4x + 2 - 6xy + 6x - 6y - 6 - 21y^2 - 42y - 21$$

$$= -14 + 3x - 33y + 2x^2 - 6xy - 21y^2$$

References